

# Algebra

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## Equalities

Mastery of [Elementary Algebra](#) (a.k.a. “College Algebra”) is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

**Theorem 0.1** (Equalities are transitive). *If  $a = b$  and  $b = c$ , then  $a = c$*

**Theorem 0.2** (Substituting equivalent expressions). *If  $a = b$ , then for any function  $f(x)$ ,  $f(a) = f(b)$*

## Inequalities

**Theorem 0.3** (Adding to both sides of an inequality). *If  $a < b$ , then  $a + c < b + c$*

**Theorem 0.4** (negating both sides of an inequality). *If  $a < b$ , then:  $-a > -b$*

**Theorem 0.5** (Multiplying both sides of an inequality by a positive number). *If  $a < b$  and  $c > 0$ , then  $ca < cb$ .*

**Theorem 0.6** (Negation is multiplication by  $-1$ ).

$$-a = (-1) * a$$

## Infimum and supremum

**Definition 0.1** (Infimum (greatest lower bound)). Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, meaning that some  $t \in \mathbb{R}$  satisfies  $t \leq a$  for all  $a \in A$ . The **infimum** of  $A$ , written  $\inf A$ , is the greatest real number  $t$  satisfying  $t \leq a$  for all  $a \in A$ :

$$\inf A \stackrel{\text{def}}{=} \max\{t \in \mathbb{R} : \forall a \in A, t \leq a\}$$

If  $A$  is nonempty but not bounded below, we write  $\inf A = -\infty$  by convention.

The maximum in Definition 0.1 always exists: that is the completeness (greatest-lower-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the infimum belongs to  $A$ , it equals the minimum:  $\inf A = \min A$ .

**Example 0.1** (Numerical examples of infimum).

- $\inf\{1, 2, 3\} = 1$ , since 1 is the smallest element.
- $\inf(0.5, 1] = 0.5 = \min[0.5, 1]$ : for intervals open below, the infimum equals the minimum of the corresponding closed-below interval, even though  $0.5 \notin (0.5, 1]$ . More generally,  $\inf(c, b] = \min[c, b] = c$  for any  $c < b$ .
- $\inf\{t \geq 0 : t > 0.5\} = 0.5$ , even though 0.5 itself is not in the set.
- $\inf\{-1, -2, -3, \dots\} = -\infty$ , because no real number is less than or equal to every element of that set.

**Definition 0.2** (Supremum (least upper bound)). Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, meaning that some  $t \in \mathbb{R}$  satisfies  $a \leq t$  for all  $a \in A$ . The **supremum** of  $A$ , written  $\sup A$ , is the smallest real number  $t$  satisfying  $a \leq t$  for all  $a \in A$ :

$$\sup A \stackrel{\text{def}}{=} \min\{t \in \mathbb{R} : \forall a \in A, a \leq t\}$$

If  $A$  is nonempty but not bounded above, we write  $\sup A = +\infty$  by convention.

The minimum in Definition 0.2 always exists: that is the completeness (least-upper-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the supremum belongs to  $A$ , it equals the maximum:  $\sup A = \max A$ .

**Example 0.2** (Numerical examples of supremum).

- $\sup\{1, 2, 3\} = 3$ , since 3 is the largest element.
- $\sup\{t \geq 0 : t < 0.5\} = 0.5$ , even though 0.5 itself is not in the set.
- $\sup\{1, 2, 3, \dots\} = +\infty$ , because no real number is greater than or equal to every element of that set.

## Sums

**Theorem 0.7** (adding zero changes nothing).

$$a + 0 = a$$

**Theorem 0.8** (Sums are symmetric).

$$a + b = b + a$$

**Theorem 0.9** (Sums are associative).

*When summing three or more terms, the order in which you sum them does not matter:*

$$(a + b) + c = a + (b + c)$$

## Products

**Theorem 0.10** (Multiplying by 1 changes nothing).

$$a \times 1 = a$$

**Theorem 0.11** (Products are symmetric).

$$a \times b = b \times a$$

**Theorem 0.12** (Products are associative).

$$(a \times b) \times c = a \times (b \times c)$$

## Division

**Theorem 0.13** (Division can be written as a product). *If  $b \neq 0$ , then*

$$\frac{a}{b} = a \times \frac{1}{b}$$

## Sums and products together

**Theorem 0.14** (Multiplication is distributive).

$$a(b + c) = ab + ac$$

**Exercise 0.1** (Expand a squared sum). Is  $(3 + 4)^2$  equal to  $3^2 + 4^2$ ? Then expand  $(a + b)^2$  for any numbers  $a$  and  $b$ , using only the distributive law and the rules above.

### Solution

*Solution 0.1.* No:  $(3 + 4)^2 = 7^2 = 49$ , while  $3^2 + 4^2 = 9 + 16 = 25$ . The difference,  $49 - 25 = 24$ , is  $2 \cdot 3 \cdot 4$ .

To see where that extra term comes from, write the square as a product and apply the distributive law (Theorem 0.14) twice:

$$\begin{aligned}(a + b)^2 &= (a + b)(a + b) && \text{(definition of a square)} \\ &= (a + b)a + (a + b)b && \text{(distributive law)} \\ &= (a^2 + ba) + (ab + b^2) && \text{(distributive law, twice)} \\ &= a^2 + ab + ab + b^2 && \text{(commutative and associative laws)} \\ &= a^2 + 2ab + b^2 && \text{(collect like terms)}\end{aligned}$$

The step “commutative and associative laws” uses Theorem 0.11 to write  $ba$  as  $ab$ , and Theorem 0.9 to drop the parentheses.

**Theorem 0.15** (Square of a sum). *For any numbers  $a$  and  $b$ ,*

$$(a + b)^2 = a^2 + 2ab + b^2$$

### Proof

*Proof.* By Solution 0.1. □

Replacing  $b$  by  $-b$  gives  $(a - b)^2 = a^2 - 2ab + b^2$ . Squared errors such as  $(y - \hat{y})^2$  are expanded this way.

## Quotients

**Definition 0.3** (Quotient). For real numbers  $a$  and  $b$  with  $b \neq 0$ , the **quotient** of  $a$  by  $b$  is the result of dividing  $a$  by  $b$ :

$$\frac{a}{b}$$

A quotient is also called a *fraction*;  $a$  is its *numerator* and  $b$  its *denominator*. A quotient whose denominator measures time or population size is often called a *rate*; in epidemiology, rates typically have such a denominator.

cf. [https://en.wikipedia.org/wiki/Rate\\_\(mathematics\)](https://en.wikipedia.org/wiki/Rate_(mathematics))

**Example 0.3** (A quotient). The quotient of 6 by 4 is  $\frac{6}{4} = 1.5$ . The quotient of 6 by 0 is undefined, because Definition 0.3 requires a nonzero denominator.

**Definition 0.4** (Ratios). A **ratio** is a quotient in which the numerator and denominator are measured using the same unit scales.

cf. <https://en.wikipedia.org/wiki/Ratio>

**Definition 0.5** (Proportion). In statistics, a **proportion** typically means a ratio where the numerator represents a subset of the denominator.

See [https://en.wikipedia.org/wiki/Population\\_proportion](https://en.wikipedia.org/wiki/Population_proportion).

See also [https://en.wikipedia.org/wiki/Proportion\\_\(mathematics\)](https://en.wikipedia.org/wiki/Proportion_(mathematics)) for other meanings.

**Definition 0.6** (Proportional). Two functions  $f(x)$  and  $g(x)$  are **proportional** if their ratio  $\frac{f(x)}{g(x)}$  does not depend on  $x$ . (cf. [https://en.wikipedia.org/wiki/Proportionality\\_\(mathematics\)](https://en.wikipedia.org/wiki/Proportionality_(mathematics)))

Additional reference for elementary algebra: [https://en.wikipedia.org/wiki/Population\\_proportion#Mathematical\\_definition](https://en.wikipedia.org/wiki/Population_proportion#Mathematical_definition)

## Exponentials and Logarithms

In these notes,  $\log\{x\}$  is the natural logarithm of  $x > 0$ , the logarithm with base  $e \approx 2.718$ , and  $\exp\{x\} = e^x$  is the exponential function. Some sources write  $\ln x$  for the natural logarithm and reserve  $\log x$  for base 10.

**Theorem 0.16** ( $\exp\{\}$  and  $\log\{\}$  are mutual inverses).

1. For every  $a > 0$ :  $\exp\{\log\{a\}\} = a$ .
2. For every  $a \in \mathbb{R}$ :  $\log\{\exp\{a\}\} = a$ .

**Theorem 0.17** (Logarithm of a product). If  $a > 0$  and  $b > 0$ , then

$$\log\{a \cdot b\} = \log\{a\} + \log\{b\}$$

**Corollary 0.1** (Logarithm of a quotient). If  $a > 0$  and  $b > 0$ , then

$$\log\left\{\frac{a}{b}\right\} = \log\{a\} - \log\{b\}$$

**i** Proof

*Proof.* Since  $a > 0$  and  $b > 0$ , the quotient  $\frac{a}{b}$  is positive, so Theorem 0.17 applies to the product  $\frac{a}{b} \cdot b$ :

$$\begin{aligned}\log\{a\} &= \log\left\{\frac{a}{b} \cdot b\right\} && (a = \frac{a}{b} \cdot b) \\ &= \log\left\{\frac{a}{b}\right\} + \log\{b\} && (\text{logarithm of a product})\end{aligned}$$

The second step applies Theorem 0.17. Subtracting  $\log\{b\}$  from both sides gives  $\log\{a\} - \log\{b\} = \log\{\frac{a}{b}\}$ . □

**Theorem 0.18** (Logarithm of a power). If  $a > 0$  and  $b \in \mathbb{R}$ , then

$$\log\{a^b\} = b \cdot \log\{a\}$$

**Theorem 0.19** (exponential of a sum).

*The exponential of a sum is equal to the product of the exponentials of the addends:*

$$\exp\{a + b\} = \exp\{a\} \cdot \exp\{b\}$$

**Corollary 0.2** (exponential of a difference).

*The exponential of a difference is the exponential of the first term divided by the exponential of the second term:*

$$\exp\{a - b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

**Theorem 0.20** (Power of a power). *If  $a > 0$  and  $b, c \in \mathbb{R}$ , then*

$$a^{bc} = (a^b)^c = (a^c)^b$$

**Example 0.4** (A negative base). With  $a = -1$ ,  $b = 2$ , and  $c = \frac{1}{2}$ :

$$\begin{aligned} a^{bc} &= (-1)^{2 \cdot \frac{1}{2}} \\ &= (-1)^1 \\ &= -1 \end{aligned}$$

but

$$\begin{aligned} (a^b)^c &= ((-1)^2)^{\frac{1}{2}} \\ &= 1^{\frac{1}{2}} \\ &= 1 \end{aligned}$$

So  $a^{bc} \neq (a^b)^c$  here, which is why Theorem 0.20 requires  $a > 0$ . The third expression,  $(a^c)^b = ((-1)^{\frac{1}{2}})^2$ , is not even a real number.

**Corollary 0.3** (natural exponential of a product).

$$\exp\{ab\} = (\exp\{a\})^b = (\exp\{b\})^a$$

**Exercise 0.2.** For  $b, c \in \mathbb{R}$ , when does  $b^c = bc$ ?

**Solution**

*Solution 0.2.* We only count a pair  $(b, c)$  when  $b^c$  is a real number, so for  $b < 0$  we only consider integer  $c$  (R agrees:  $(-8)^{(1/3)}$  is NaN). With that convention,  $bc = b^c$  in each of the following cases:

1.  $c = 1$ , for every  $b$ .
2.  $b = 0$  and  $c > 0$ , since then  $b^c = 0 = bc$ . ( $b = 0$  and  $c = 0$  fails, since  $0^0 = 1 \neq 0$ .)
3.  $b > 0$ ,  $c > 0$ ,  $c \neq 1$ , and  $b = \exp\left\{\frac{\log\{c\}}{c-1}\right\}$ . For  $b > 0$ , dividing both sides of  $b^c = bc$  by  $b$  gives  $b^{c-1} = c$ , which needs  $c > 0$  because  $b^{c-1} > 0$ ; taking logarithms then gives  $(c-1)\log\{b\} = \log\{c\}$ . For example,  $c = 2$  gives  $b = 2$ , and indeed  $2^2 = 4 = 2 \cdot 2$ .

4.  $b < 0$  and  $c$  is an odd integer with  $c \geq 3$ , with  $b = -c^{1/(c-1)}$ ; for example,  $b = -\sqrt{3}$  and  $c = 3$  give  $b^c = -3\sqrt{3} = bc$ .
5.  $b < 0$  and  $c$  is an even integer with  $c \leq -2$ , with  $b = -(-c)^{1/(c-1)}$ ; for example,  $b = -2^{-1/3}$  and  $c = -2$  give  $b^c = 2^{2/3} = bc$ .

For  $b < 0$ , cases 4 and 5 come from  $b^{c-1} = c$  as well: when  $c - 1$  is even,  $b^{c-1} > 0$ , so  $c$  must be positive; when  $c - 1$  is odd,  $b^{c-1} < 0$ , so  $c$  must be negative.

See the red contours in Figure 2 for a visualization of the  $b \geq 0$  cases.

*(This 3D surface is interactive; see the HTML or slide version of this page to view it.)*

Figure 1: Graph of  $b * c$  and  $b^c$

*(This 3D surface is interactive; see the HTML or slide version of this page to view it.)*

Figure 2: **Graph of  $b^c - b * c$ .** Red contour lines show where  $b^c = b * c$ .

**Exercise 0.3.** For  $a \geq 0$ ,  $b, c \in \mathbb{R}$ , when does  $(a^b)^c = a^{(b^c)}$ ?

#### Solution

*Solution 0.3.* Short answer: rarely (that's all you need to know for this course).

Long answer:

Split on whether  $a > 0$  or  $a = 0$ , because the logarithm we use for  $a > 0$  is undefined at  $a = 0$ .

**Case  $a > 0$ .** By Theorem 0.20,  $(a^b)^c = a^{bc}$ , so the question becomes when  $a^{bc} = a^{(b^c)}$  (for pairs  $(b, c)$  where  $b^c$  is defined). Because  $a > 0$ , both sides are positive, and we can take logarithms (Theorem 0.18):

$$\begin{aligned}
 a^{bc} &= a^{(b^c)} \\
 \log\{a^{bc}\} &= \log\{a^{(b^c)}\} \quad (\text{take logarithms of both sides}) \\
 bc \cdot \log\{a\} &= b^c \cdot \log\{a\} \quad (\text{logarithm of a power})
 \end{aligned} \tag{1}$$

The last line of Equation 1 holds exactly when

1.  $a = 1$  (so that  $\log\{a\} = 0$ ), or
2.  $bc = b^c$  (see Exercise 0.2).

**Case  $a = 0$ .** Here we cannot take logarithms, so we work from the values of powers of 0:  $0^s = 0$  for  $s > 0$ ,  $0^0 = 1$ , and  $0^s$  is undefined for  $s < 0$ .

- If  $b < 0$ , then  $0^b$  is undefined, so  $(a^b)^c$  is undefined.
- If  $b > 0$ , then  $(0^b)^c = 0^c$  and  $b^c > 0$ , so  $0^{(b^c)} = 0$ ; the two sides agree exactly when  $c > 0$ .
- If  $b = 0$ , then  $(0^0)^c = 1^c = 1$ ; for  $c > 0$ ,  $0^{(0^c)} = 0^0 = 1$ , so the two sides agree, and for  $c \leq 0$  they do not.

So for  $a = 0$ ,  $(a^b)^c = a^{(b^c)}$  exactly when  $b \geq 0$  and  $c > 0$ .

In particular, when  $a = 0$ ,  $b \geq 0$ , and  $c = 0$ , the two sides differ:

$$\begin{aligned}(a^b)^c &= (0^b)^0 \\ &= 1\end{aligned}$$

$$\begin{aligned}a^{(b^c)} &= 0^{(b^0)} \\ &= 0^1 \\ &= 0\end{aligned}$$

Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.